120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 9: Supervised Learning

Day 58: Gradient Descent Mathematics

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1. Optimization Objective

Given an empirical risk:

$$J(\theta) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(y_i, f(x_i; \theta))$$

the learning problem becomes:

$$\theta^* = \arg\min_{\theta} J(\theta)$$

When no closed-form solution exists, we rely on iterative optimization.

2. Gradient and Directional Derivatives

Let:

$$\theta = (\theta_1, \ldots, \theta_p)$$

The gradient vector is:

$$\nabla J(\theta) = \left(\frac{\partial J}{\partial \theta_1}, \ldots, \frac{\partial J}{\partial \theta_p} \right)$$

It points in the direction of steepest ascent.
Thus, to minimize $J(\theta)$, we move in the opposite direction.

3. Gradient Descent Update Rule

The iterative update is:

$$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t)$$

where $\eta > 0$ is the learning rate.

4. Taylor Expansion Justification

Using first-order Taylor approximation:

$$J(\theta + \Delta\theta) \approx J(\theta) + \nabla J(\theta)^T \Delta\theta$$

If we set:

$$\Delta\theta = -\eta \nabla J(\theta)$$

then for sufficiently small η:

$$J(\theta + \Delta\theta) < J(\theta)$$

Thus, each step reduces the objective function.

5. Example: Linear Regression

For:

$$f(x_i; \theta) = \theta^T x_i$$

MSE objective:

$$J(\theta) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \theta^T x_i)^2$$

Gradient:

$$\nabla J(\theta) = -\frac{2}{n} X^T (y - X\theta)$$

Update rule:

$$\theta_{t+1} = \theta_t + \frac{2\eta}{n} X^T (y - X\theta_t)$$

6. Convergence Conditions

If $J(\theta)$ is convex and its gradient is Lipschitz continuous, then:

$$0 < \eta < \frac{2}{L}$$

guarantees convergence to the global minimum.

Conclusion

Gradient descent is a first-order optimization method that iteratively updates parameters in the direction of steepest descent. It forms the foundation of modern machine learning and deep learning systems.